Campus Services

Information Technology

Declining/Inclining Balance POS Interface Requirements

Version 1.0

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1.0 Overview

1.1 Background

Introduction:

- This document contains the requirements for interfacing Declining/Inclining balance, and Inquiries into the ARAMARK POS Interface.
- This interface specifications are provided with this document. Together they will provide the information necessary to complete an interface to the ARAMARK POS systems utilized by many of ARAMARK's Clients.
- This interface will be designed to work over the standard TCP/IP Port 80.
- XML will be used as the format for data streams going between source systems, and the POS relay systems.
- To the extent possible, standard XML schemas will be utilized
- Triple DES will be used to Encrypt the data.

1.2 Current State

Brief:

- Campus Services utilizes several major POS systems. Two of these systems are operated by and mainted by Campus IT. These applications all utilize their own proprietary communications protocols by which external applications can interface with them. This document, and the attached interface specification is meant to bridge the interoperability gaps of the two POS systems, and reduce the complexity of integrating with ARAMARK systems

1.3 Objectives

Brief:

- Provide a common standards based approach to interfacing external systems with the POS systems utilized in the ARAMARK Campus Services environment.
- Provide flexibility and lower startup costs through the use of a well defined interface.
- Enable a secure process via the use of commonly available security tools.

1.4 Assumptions

Brief:

- Communications with the ARAMARK POS Interface can will go over the standard port: 80

1.6 Approach

Brief:

- Utilize a standards based framework. This includes the use of web services, and transactions based on the ARTS standards.
- Test via ARAMARK test server in Philadelphia

1.7 Scope

In Scope:

- ARAMARK has developed web services to support the following transaction types.
    - Inquiry
    - Inclining/Declining Balance
- The following transactions also exist, but are not a part of this implementation.
    - Cash Equivalency
    - Cash/Credit
    - Board Meals
    - Transaction Summary consisting of transaction totals, tenders, item detail, discounts, etc….

2.0 High-Level Functional Requirements

2.1 Overview

ARAMARK has developed specifications for a web services based interface that will be utilized as a hub to broker transactions between external sources and the standard POS system utilized on campus dining facilities managed by ARAMARK.

This interface will be developed as a Secure web service in Visual Studio .Net 2003.

The interface will utilize a basic framework for POS transactions developed by ARTS, the Association for Retail Technology Standards. The ARTS home page is located at http://www.nrf-arts.org . ARTS has developed a set of XML schemas to standardize communications between systems that handle POS data. These schemas cover areas from Tenders, to item detail, to discounts, and taxes.

ARAMARK has developed some extensions to that framework to handle University specific transactions like Declining Balance, Board Plans, and Cash Equivalency transactions.

The details of the interfaces for all transactions can be located in the ARAMARK Campus Services POS Interface document.

2.2 System Configuration

2.2.1 Configuration

Every call to the web service must have a client and device identifier. This is a unique value which identifies to the web service what school, and device the transaction is originating from. These will be mapped to appropriate devices in the pos system the transaction is being sent to.

In addition, the PC originating the request will have to know the IP address, or DNS name of web service.

2.3 Supported Transactions

2.3.2 Inquiry

Inquiry Transactions are utilized to request the balance a patron has on the POS system without actually modifying any funds. The request sent to the POS system includes the card number of a patron. The POS interface will reply with the balance available to the patron. If the patron has a balance available, they can spend up to that amount. It is possible for a patron to spend beyond the balance on the card. If this is the case, the system will show that the user has a credit balance.

- An inquiry should be performed as the first step in performing a declining balance which will be described next.
- Inquiries may also be performed in a standalone manner to determine the available balance of the patron

2.3.2 Declining Balance

Declining Balance involves the use of available funds from an account on one of the ARAMARK POS systems. Declining Balance is generally a two step process Part one of the process is to perform an inquiry transaction as described in section 2.3.1. The system will return with the available balance for the patron. If the patron has sufficient funds, a declining balance transaction is then sent to remove those funds from the patron's account.

- If the student does not have sufficient funds to cover the whole transaction, then the portion of the transaction indicated by the balance returned from the inquiry should be sent as the declining balance amount. The originating system should then prompt the user for another payment for the remaining balance on the transaction
- While the patron has funds available, the inquiry will indicate this amount. If the patron has 0, or less money, then the system will indicate if the student can continue to spend money in a credit mode.
- Some schools work exclusively in a credit or inclining balance. This is done with payroll deduction as well, where we create a monthly statement on the POS system. In this case, the POS system will always indicate that the patron has a credit balance.
- The use of declining balance generally has tax implications. The Inquiry transaction to the POS system will return a tax flag indicating whether items charged to that DB account should be taxed or not. If items were taxable, but the flag indicates not to use taxes, then the tax amount should be removed prior to processing the transaction.

2.3.3 Credit to Account

Credit to Account is not a credit card transaction. Credit to Account is used add money back to a Declining Balance account.

- The ARAMARK interface will have a configuration to determine whether Credit Account Transactions are allowed for a given device.

2.5 Operator Procedures

2.5.1 Overview

Appropriate procedures should exist in the host system for managing processes required by the addition of the communication mechanism, and transaction processing described in this document. This includes configuring the register with data required by the POS Interface,

2.5.2 Configuration

Ideally, there should be a place within the host software to configure settings required for communications with the ARAMARK web service.

2.5.3 Reporting

Reporting should be available in the host system to reconcile inclining and declining balance transactions against the ARAMARK POS system. An equivalent report would run on the ARAMARK POS system to detail the same transactions for the same time period. Any discrepancies should be researched and addressed.

3.0 Appendix

3.1 POS Interface Specifications

See attached documents

3.2 POS Interface Error Codes

See attached documents